

Algebra First-Year Exam, January 2019, Part 1

Answer four problems. (If you turn in more, the first four will be graded.) Within reason, you may quote theorems as long as you state them clearly.

1. State and prove Sylow's First Theorem.
2. Consider the symmetric group S_6 , and the corresponding alternating group A_6 .
 - (a) How many elements of order 4 are there in S_6 ? Justify your answer.
 - (b) How many elements of order 4 are there in A_6 ? Justify your answer.
3. Let G be the alternating group A_{100} , and assume G acts on a finite set Ω .
 - (a) Prove that, if $|\Omega| = 90$, then G acts trivially on Ω , i.e. G has 90 orbits of size 1 on Ω .
 - (b) Prove that if $|\Omega| = 120$ and G does not act trivially on Ω , then G has, at most, 21 orbits on Ω .
4. Prove that every group of order 30 has a normal subgroup of order 15.
5. Let G be a group, let N be a normal subgroup of G , and let H be a subgroup of G . Assume that H is solvable. Prove that HN/N is a solvable group.
6. Let G be a group of order 39. Does G need to be abelian? Prove that your answer is correct.